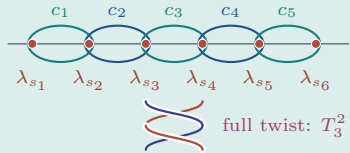


the curve D_a , a of order two

$$y^2 = \prod_{i \in S} (t - \lambda_i), \quad |S| = 2g + 2 = 6$$



$$T_k(z) = z + \langle z, c_k \rangle c_k$$

$$\langle c_k, c_{k \pm 1} \rangle = \pm 1, \text{ the others } 0$$

the chain generates $\mathfrak{sp}(V_a)$



$$N_k = \langle \cdot, c_k \rangle c_k = \frac{1}{2} c_k^2 \text{ in } \text{Sym}^2 V_a$$

$$[c_k^2, c_{k+1}^2] = \pm 4 c_k c_{k+1}$$

$$c_i c_{k+1} = \pm \frac{1}{2} [c_i c_k, c_{k+1}^2] \text{ for } i < k$$

so the $c_i c_j$ span $\text{Sym}^2 V_a = \mathfrak{sp}(V_a)$,
of dimension $g(2g + 1)$;

a power of each T_k^2 lies in Π' , so

$\overline{\Pi'} \supset \text{Sp}(V_a)$ (Zariski closure)

order two

$\text{Sp}(V_a)$ -invariants of $\bigwedge^r V_a$:
the multiples of $\omega_a^{r/2}$, the image
of $\theta_a^{r/2}$; so $x_{[a]}$ is algebraic

order three

Achter–Pries: SU , acting as $\text{SL}(V_a)$;
a class only if $r = \dim V_a$, $p_a = r/2$;
Weil type for $\mathbb{Q}(\sqrt{-3})$, δ trivial;
algebraic for $r \leq 6$ by Markman
HC: $d = 2$, every r ; $d = 3$, $r \leq 6$